PROGRAM 1.1 — Student Information System

```
class StuinfoSys {

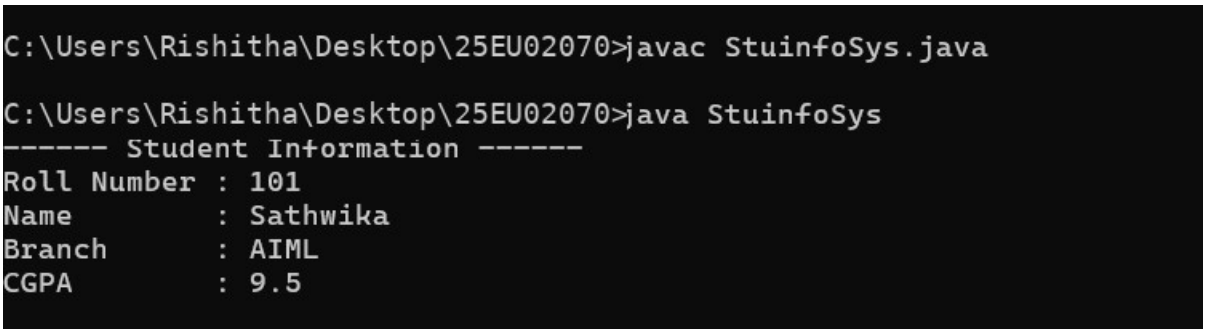
   int rollNo;
   String name;
   String branch;
   double cgpa;

   void setData(int r, String n, String b, double c) {
      rollNo = r;
      name = n;
      branch = b;
      cgpa = c;
   }

   void display() {
      System.out.println("------ Student Information ------");
      System.out.println("Roll Number : " + rollNo);
      System.out.println("Name        : " + name);
      System.out.println("Branch      : " + branch);
      System.out.println("CGPA        : " + cgpa);
   }

   public static void main(String[] args) {
      StuinfoSys s = new StuinfoSys();
      s.setData(101, "Sathwika", "AIML", 9.5);
      s.display();
   }
}
```

OUTPUT:



PROGRAM 1.2 — Employee Information System

```
class EmpinfoSys {

    int empId;
    String name;
    String department;
    double salary;

    void setData(int id, String n, String d, double s) {
        empId = id;
        name = n;
        department = d;
        salary = s;
    }

    void display() {
        System.out.println("------ Employee Information ------");
        System.out.println("Employee ID : " + empId);
        System.out.println("Name        : " + name);
        System.out.println("Department  : " + department);
        System.out.println("Salary      : " + salary);
    }

    public static void main(String[] args) {
        EmpinfoSys e = new EmpinfoSys();
        e.setData(1001, "Rahul", "HR", 35000);
        e.display();
```

```
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac EmpinfoSys.java

C:\Users\Rishitha\Desktop\25EU02070>java EmpinfoSys
------ Employee Information ------
Employee ID : 1001
Name        : Rahul
Department  : HR
Salary      : 35000.0
```

PROGRAM 2.1 — Book Details

```
class BookDetails {

    int bookId;
    String title;
    String author;

    BookDetails(int id, String t, String a) {
        bookId = id;
        title = t;
        author = a;
    }

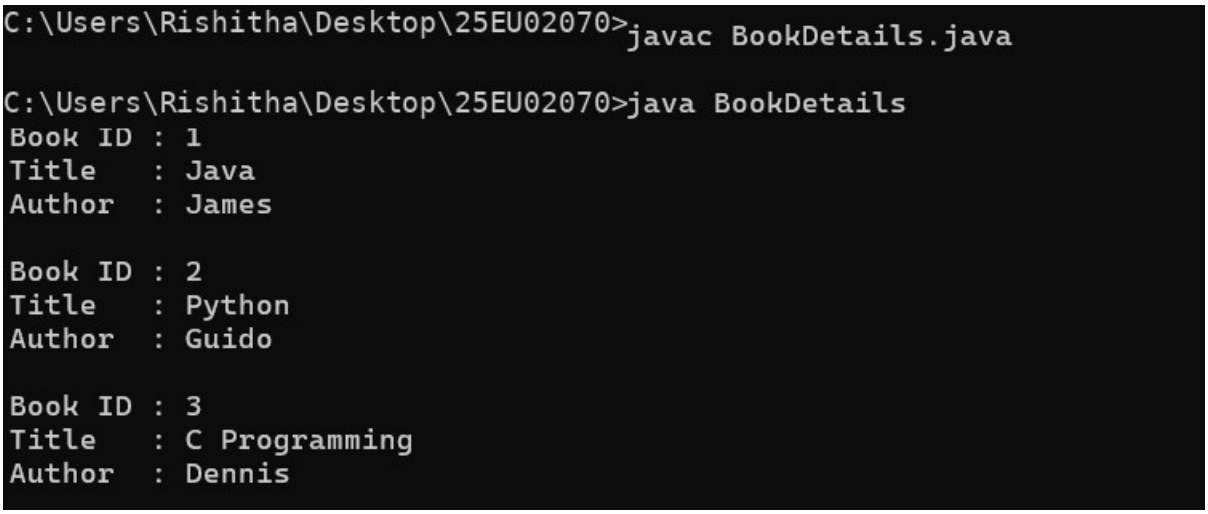
    void display() {
        System.out.println("Book ID : " + bookId);
        System.out.println("Title   : " + title);
        System.out.println("Author  : " + author);
        System.out.println();
    }

    public static void main(String[] args) {
        BookDetails b1 = new BookDetails(1, "Java", "James");
        BookDetails b2 = new BookDetails(2, "Python", "Guido");
        BookDetails b3 = new BookDetails(3, "C Programming", "Dennis");

        b1.display();
        b2.display();
        b3.display();
```

```
    }
}
```

OUTPUT:



PROGRAM 2.2 — Array of Student Objects

```
import java.util.Scanner;

class StudentArray {

    int rollNo;
    String name;

    void input(Scanner sc) {
        System.out.print("Enter Roll No: ");
        rollNo = sc.nextInt();

        System.out.print("Enter Name: ");
        name = sc.next();
    }

    void display() {
        System.out.println(rollNo + " " + name);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        StudentArray[] s = new StudentArray[5];

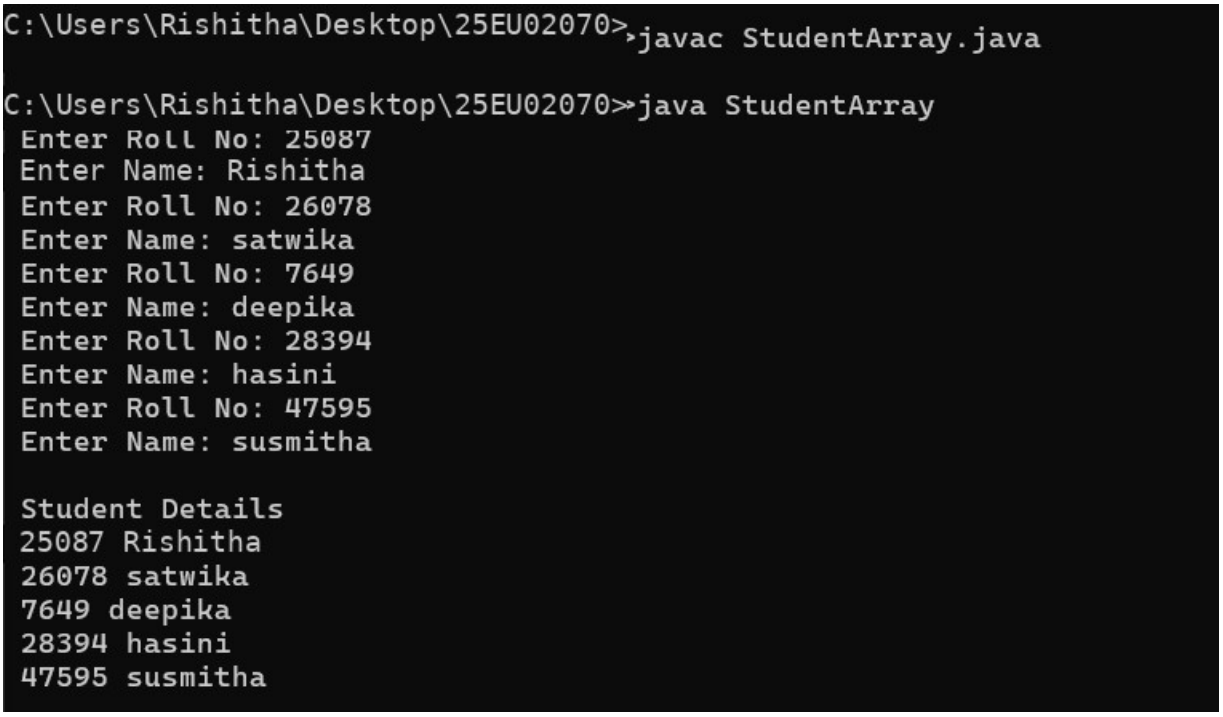
        for (int i = 0; i < 5; i++) {
            s[i] = new StudentArray();
            s[i].input(sc);
        }

        System.out.println("\nStudent Details");

        for (int i = 0; i < 5; i++) {
            s[i].display();
        }

        sc.close();
    }
}
```

OUTPUT:



PROGRAM 3.1 — Reference Assignment

```
class RefAssignment {

   String name;

   public static void main(String[] args) {
      RefAssignment s1 = new RefAssignment();

      s1.name = "Ravi";

      RefAssignment s2 = s1;

      s2.name = "Sathwika";

      System.out.println("s1 Name : " + s1.name);
      System.out.println("s2 Name : " + s2.name);
   }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac EmpReference.java

C:\Users\Rishitha\Desktop\25EU02070>java EmpReference
 Are both references same? true
```

PROGRAM 3.2 — Comparing Object References

```
class EmpReference {

   String name;

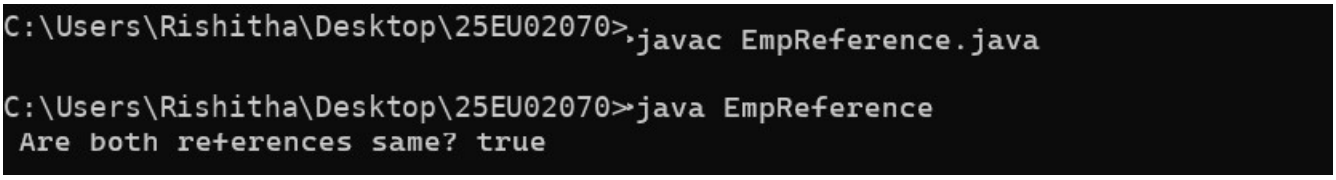
   public static void main(String[] args) {
      EmpReference e1 = new EmpReference();

      e1.name = "Anil";

      EmpReference e2 = e1;
```

```
        System.out.println("Are both references same? " + (e1 == e2));
    }
}
```

OUTPUT:



PROGRAM 4.1 — Calculator Using Methods

```
class CalcMethods {

    int add(int a, int b) {
        return a + b;
    }

    int subtract(int a, int b) {
        return a - b;
    }

    int multiply(int a, int b) {
        return a * b;
    }

    double divide(int a, int b) {
        return (double) a / b;
    }

    public static void main(String[] args) {
        CalcMethods c = new CalcMethods();

        System.out.println("Addition       = " + c.add(10, 5));
        System.out.println("Subtraction    = " + c.subtract(10, 5));
        System.out.println("Multiplication = " + c.multiply(10, 5));
        System.out.println("Division       = " + c.divide(10, 5));
```

```
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac CalcMethods.java

C:\Users\Rishitha\Desktop\25EU02070>java CalcMethods
Addition       = 15
Subtraction    = 5
Multiplication = 50
Division       = 2.0
```

PROGRAM 4.2 — Rectangle Operations

```
class RectOperations {

    double length;
    double width;

    RectOperations(double l, double w) {
        length = l;
        width = w;
    }

    double area() {
        return length * width;
    }

    double perimeter() {
        return 2 * (length + width);
    }

    public static void main(String[] args) {
        RectOperations r = new RectOperations(10, 5);

        System.out.println("Area      = " + r.area());
        System.out.println("Perimeter = " + r.perimeter());
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac RectOperations.java

C:\Users\Rishitha\Desktop\25EU02070>java RectOperations
Area      = 50.0
Perimeter = 30.0
```

PROGRAM 5.1 — Student Constructors

```
class StudentConstructors {

   int rollNo;
   String name;
   String branch;

   StudentConstructors() {
      rollNo = 101;
      name = "Sathwika";
      branch = "AIML";
   }

   StudentConstructors(int r, String n, String b) {
      rollNo = r;
      name = n;
      branch = b;
   }

   void display() {
      System.out.println("Roll Number : " + rollNo);
      System.out.println("Name       : " + name);
      System.out.println("Branch     : " + branch);
      System.out.println();
   }

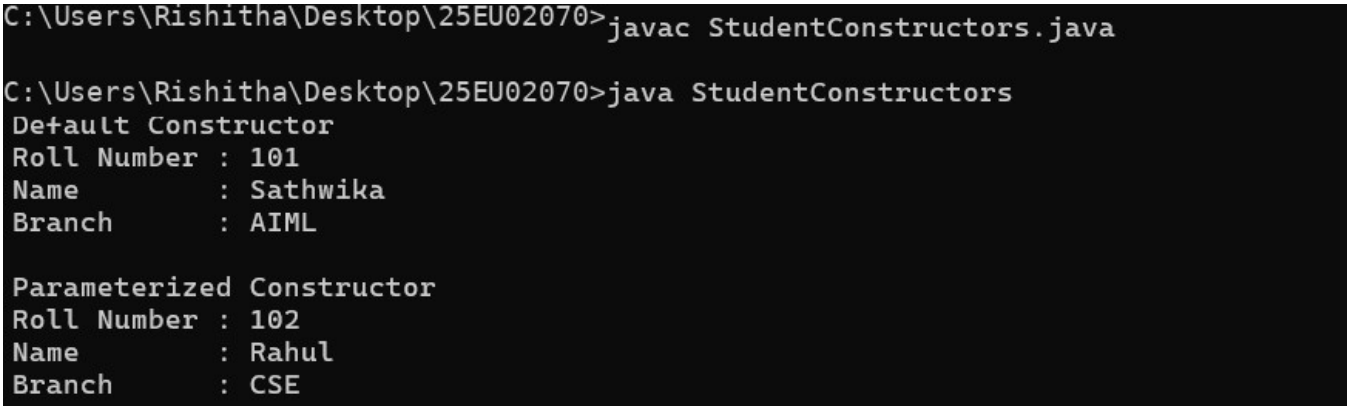
   public static void main(String[] args) {
      StudentConstructors s1 = new StudentConstructors();
      StudentConstructors s2 = new StudentConstructors(102, "Rahul", "CSE");

      System.out.println("Default Constructor");
```

```
        s1.display();

        System.out.println("Parameterized Constructor");
        s2.display();
    }
}
```

OUTPUT:



PROGRAM 5.2 — Constructor Overloading

```
class ConstructorOverload {

    int accountNumber;
    String accountHolder;
    double balance;

    ConstructorOverload() {
        accountNumber = 1001;
        accountHolder = "Sathwika";
        balance = 0;
    }

    ConstructorOverload(int accNo, String holder) {
        accountNumber = accNo;
        accountHolder = holder;
        balance = 0;
    }
```

```
    ConstructorOverload(int accNo, String holder, double bal) {
        accountNumber = accNo;
        accountHolder = holder;
        balance = bal;
    }

    void display() {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Account Holder : " + accountHolder);
        System.out.println("Balance        : " + balance);
        System.out.println();
    }

    public static void main(String[] args) {
        ConstructorOverload b1 = new ConstructorOverload();
        ConstructorOverload b2 = new ConstructorOverload(1002, "Rahul");
        ConstructorOverload b3 = new ConstructorOverload(1003, "Anil", 5000);

        b1.display();
        b2.display();
        b3.display();
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac ConstructorOverload.java

C:\Users\Rishitha\Desktop\25EU02070>java ConstructorOverload
 Account Number : 1001
 Account Holder : Sathwika
 Balance        : 0.0

 Account Number : 1002
 Account Holder : Rahul
 Balance        : 0.0

 Account Number : 1003
 Account Holder : Anil
 Balance        : 5000.0
```

PROGRAM 6.1 — Using this Keyword

```
class ThisKeywordDemo {

   int rollNo;
   String name;

   ThisKeywordDemo(int rollNo, String name) {
      this.rollNo = rollNo;
      this.name = name;
   }

   void display() {
      System.out.println("Roll Number : " + rollNo);
      System.out.println("Name       : " + name);
   }

   public static void main(String[] args) {
      ThisKeywordDemo s = new ThisKeywordDemo(101, "Sathwika");
      s.display();
   }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac ThisKeywordDemo.java

C:\Users\Rishitha\Desktop\25EU02070>java ThisKeywordDemo
Roll Number : 101
Name        : Sathwika
```

PROGRAM 6.2 — Constructor Chaining

```
class ConstructorChaining {

   ConstructorChaining() {
      this(101);
      System.out.println("Default Constructor");
   }
```

```
    ConstructorChaining(int rollNo) {
        this(rollNo, "Sathwika");
        System.out.println("Roll Number : " + rollNo);
    }

    ConstructorChaining(int rollNo, String name) {
        System.out.println("Roll Number : " + rollNo);
        System.out.println("Name        : " + name);
    }

    public static void main(String[] args) {
        new ConstructorChaining();
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac ConstructorChaining.java

C:\Users\Rishitha\Desktop\25EU02070>java ConstructorChaining
 Roll Number : 101
 Name        : Sathwika
 Roll Number : 101
 Default Constructor
```

PROGRAM 7.1 — Demonstrating Garbage Collection

```
class GarbageCollection {

    public static void main(String[] args) {

        GarbageCollection obj1 = new GarbageCollection();
        GarbageCollection obj2 = new GarbageCollection();

        obj1 = null;
        obj2 = null;

        System.gc();

        System.out.println("Garbage Collection Requested");
```

```
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac GarbageCollection.java

C:\Users\Rishitha\Desktop\25EU02070>java GarbageCollection
 Garbage Collection Requested
```

PROGRAM 7.2 — Object Eligibility for Garbage Collection

```
class GarbageEligibility {

    public static void main(String[] args) {

        GarbageEligibility obj1 = new GarbageEligibility();
        GarbageEligibility obj2 = new GarbageEligibility();

        // obj1 becomes eligible for garbage collection
        obj1 = null;

        // The old object referenced by obj2
        // becomes eligible for garbage collection
        obj2 = new GarbageEligibility();

        // Anonymous object becomes eligible
        new GarbageEligibility();

        System.gc();

        System.out.println("Objects are eligible for Garbage Collection");
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac GarbageEligibility.java

C:\Users\Rishitha\Desktop\25EU02070>java GarbageEligibility
Objects are eligible for Garbage Collection
```

PROGRAM 8.1 — Method Overloading

```
class ArithmeticOverload {

   int add(int a, int b) {
      return a + b;
   }

   int add(int a, int b, int c) {
      return a + b + c;
   }

   double add(double a, double b) {
      return a + b;
   }

   public static void main(String[] args) {
      ArithmeticOverload obj = new ArithmeticOverload();

      System.out.println("Sum = " + obj.add(10, 20));
      System.out.println("Sum = " + obj.add(10, 20, 30));
      System.out.println("Sum = " + obj.add(10.5, 20.5));
   }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac ArithmeticOverload.java

C:\Users\Rishitha\Desktop\25EU02070>java ArithmeticOverload
Sum = 30
Sum = 60
Sum = 31.0
```

PROGRAM 8.2 — Overloading Area Methods

```
class AreaOverload {

   double area(double radius) {
      return 3.14 * radius * radius;
```

```
    }

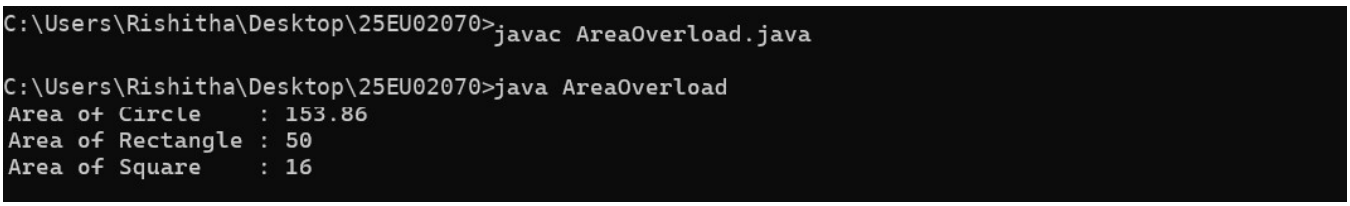
    int area(int length, int breadth) {
        return length * breadth;
    }

    int area(int side) {
        return side * side;
    }

    public static void main(String[] args) {
        AreaOverload obj = new AreaOverload();

        System.out.println("Area of Circle    : " + obj.area(7.0));
        System.out.println("Area of Rectangle : " + obj.area(10, 5));
        System.out.println("Area of Square    : " + obj.area(4));
    }
}
```

OUTPUT:



PROGRAM 9.1 — Passing Student Object

```
class PassingStudentObj {

   int rollNo;
   String name;
   String branch;

   PassingStudentObj(int rollNo, String name, String branch) {
      this.rollNo = rollNo;
      this.name = name;
      this.branch = branch;
   }

   static void displayStudent(PassingStudentObj s) {
      System.out.println("Roll Number : " + s.rollNo);
      System.out.println("Name       : " + s.name);
      System.out.println("Branch     : " + s.branch);
   }

   public static void main(String[] args) {

      PassingStudentObj student =
         new PassingStudentObj(101, "Sathwika", "AIML");

      displayStudent(student);
   }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac PassingStudentObj.java

C:\Users\Rishitha\Desktop\25EU02070>java PassingStudentObj
Roll Number : 101
Name        : Sathwika
Branch      : AIML
```

PROGRAM 9.2 — Comparing Employee Salaries

```
class CompareEmpSalary {

   int empId;
   String name;
   double salary;

   CompareEmpSalary(int empId, String name, double salary) {
      this.empId = empId;
      this.name = name;
      this.salary = salary;
   }

   static void compareSalary(
      CompareEmpSalary e1,
      CompareEmpSalary e2) {

      if (e1.salary > e2.salary) {
         System.out.println("Higher Salary Employee : " + e1.name);
         System.out.println("Salary : " + e1.salary);
      } else if (e2.salary > e1.salary) {
         System.out.println("Higher Salary Employee : " + e2.name);
         System.out.println("Salary : " + e2.salary);
      } else {
         System.out.println("Both employees have equal salary.");
      }
   }

   public static void main(String[] args) {

      CompareEmpSalary e1 =
         new CompareEmpSalary(101, "Rahul", 35000);

      CompareEmpSalary e2 =
         new CompareEmpSalary(102, "Anil", 45000);

      compareSalary(e1, e2);
```

```
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac CompareEmpSalary.java

C:\Users\Rishitha\Desktop\25EU02070>java CompareEmpSalary
Higher Salary Employee : Anil
Salary : 45000.0
```

PROGRAM 10.1 — Returning a Student Object

```
class ReturnStudentObj {

    int rollNo;
    String name;
    String branch;

    ReturnStudentObj(int rollNo, String name, String branch) {
        this.rollNo = rollNo;
        this.name = name;
        this.branch = branch;
    }

    static ReturnStudentObj createStudent() {
        ReturnStudentObj s =
            new ReturnStudentObj(101, "Sathwika", "AIML");
        return s;
    }

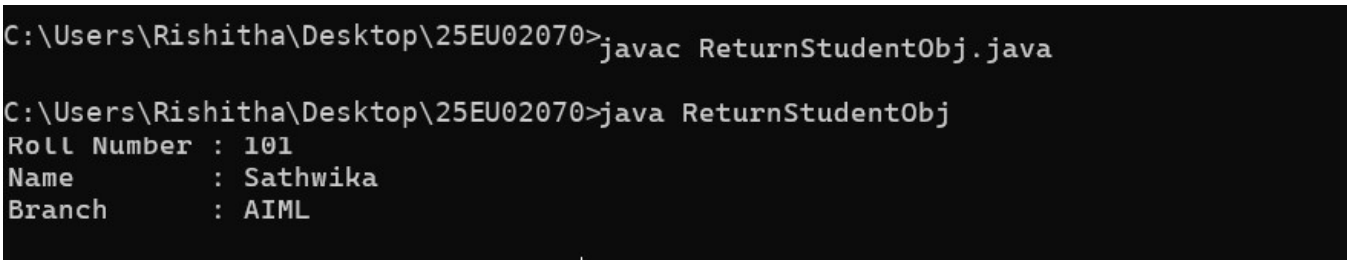
    void display() {
        System.out.println("Roll Number : " + rollNo);
        System.out.println("Name       : " + name);
        System.out.println("Branch     : " + branch);
    }

    public static void main(String[] args) {

        ReturnStudentObj student = createStudent();
        student.display();
```

```
    }
}
```

OUTPUT:



PROGRAM 10.2 — Returning a Bank Account Object

```
class ReturnBankObj {

    int accountNumber;
    String accountHolder;
    double balance;

    ReturnBankObj(int accountNumber, String accountHolder, double balance) {
        this.accountNumber = accountNumber;
        this.accountHolder = accountHolder;
        this.balance = balance;
    }

    static ReturnBankObj updateBalance(
        ReturnBankObj account,
        double amount) {

        account.balance = account.balance + amount;
        return account;
    }

    void display() {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Account Holder : " + accountHolder);
        System.out.println("Balance        : " + balance);
    }
```

```
    public static void main(String[] args) {

        ReturnBankObj account =
            new ReturnBankObj(1001, "Sathwika", 5000);

        account = updateBalance(account, 2000);
        account.display();
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac ReturnBankObj.java

C:\Users\Rishitha\Desktop\25EU02070>java ReturnBankObj
Account Number : 1001
Account Holder : Sathwika
Balance        : 7000.0
```

PROGRAM 11.1 — Static Variable

```
class StaticStudentCount {

    int rollNo;
    String name;

    static int count = 0;

    StaticStudentCount(int rollNo, String name) {
        this.rollNo = rollNo;
        this.name = name;
        count++;
    }

    void display() {
        System.out.println("Roll Number : " + rollNo);
        System.out.println("Name        : " + name);
    }
```

```
    public static void main(String[] args) {

        StaticStudentCount s1 =
            new StaticStudentCount(101, "Sathwika");

        StaticStudentCount s2 =
            new StaticStudentCount(102, "Rahul");

        StaticStudentCount s3 =
            new StaticStudentCount(103, "Anil");

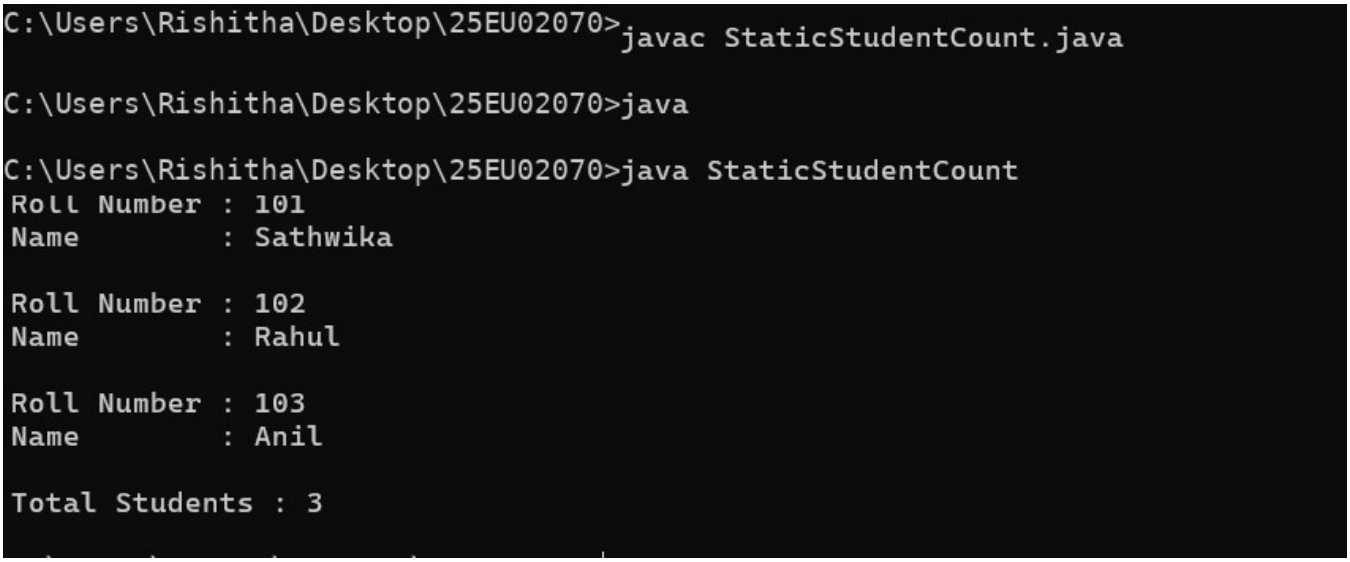
        s1.display();
        System.out.println();

        s2.display();
        System.out.println();

        s3.display();

        System.out.println("
Total Students : " + count);
    }
}
```

OUTPUT:



PROGRAM 11.2 — Static Methods

```
class StaticUtility {
  static int square(int n) {
      return n * n;
   }

   static int cube(int n) {
      return n * n * n;
   }

   static long factorial(int n) {

      long fact = 1;

      for (int i = 1; i <= n; i++) {
         fact = fact * i;
      }

      return fact;
   }

   public static void main(String[] args) {

      int n = 5;

      System.out.println("Square    : " + square(n));
      System.out.println("Cube      : " + cube(n));
      System.out.println("Factorial : " + factorial(n));
   }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac StaticUtility.java

C:\Users\Rishitha\Desktop\25EU02070>java StaticUtility
Square    : 25
Cube      : 125
Factorial  : 120
```

PROGRAM 12.1 — Final Keyword

```
class FinalKeywordDemo {

   final int MAX_MARKS = 100;

   final void display() {

      System.out.println("Final method executed.");
      System.out.println("Maximum Marks : " + MAX_MARKS);
   }

   final class FinalClass {

      void show() {
         System.out.println("This is a final class.");
      }
   }public static void main(String[] args) {

      FinalKeywordDemo obj = new FinalKeywordDemo();

      obj.display();

      FinalKeywordDemo.FinalClass f =
         obj.new FinalClass();

      f.show();
   }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac FinalKeywordDemo.java

C:\Users\Rishitha\Desktop\25EU02070>java FinalKeywordDemo
Final method executed.
Maximum Marks : 100
This is a final class.
```

PROGRAM 12.2 — Blank Final Variable

```
class BlankFinalEmp {
   final int employeeId;
   String name;

   BlankFinalEmp(int employeeId, String name) {
      this.employeeId = employeeId;
      this.name = name;
   }

   void display() {
      System.out.println("Employee ID : " + employeeId);
      System.out.println("Name        : " + name);
   }

   public static void main(String[] args) {

      BlankFinalEmp e1 =
         new BlankFinalEmp(1001, "Sathwika");

      BlankFinalEmp e2 =
         new BlankFinalEmp(1002, "Rahul");

      e1.display();
System.out.println();

      e2.display();
   }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac BlankFinalEmp.java

C:\Users\Rishitha\Desktop\25EU02070>java BlankFinalEmp
Employee ID : 1001
Name        : Sathwika

Employee ID : 1002
Name        : Rahul
```

PROGRAM 13.1 — College and Department

```
class CollegeDeptNested {

    String collegeName = "ABC Engineering College";

    static class Department {

        String departmentName;
        String hodName;

        Department(String departmentName, String hodName) {
            this.departmentName = departmentName;
            this.hodName = hodName;
        }

        void display() {
            System.out.println("Department : " + departmentName);
            System.out.println("HOD        : " + hodName);
        }
    }

    public static void main(String[] args) {

        CollegeDeptNested college =
            new CollegeDeptNested();

        CollegeDeptNested.Department dept =
            new CollegeDeptNested.Department("AIML", "Dr. Kumar");

        System.out.println("College : " + college.collegeName);

        dept.display();
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac CollegeDeptNested.java

C:\Users\Rishitha\Desktop\25EU02070>java CollegeDeptNested
College : ABC Engineering College
Department : AIML
HOD        : Dr. Kumar
```

PROGRAM 13.2 — Employee Address

```
class EmployeeAddressNested {

   String employeeName;
   int employeeId;

   EmployeeAddressNested(String employeeName, int employeeId) {
      this.employeeName = employeeName;
      this.employeeId = employeeId;
   }

   static class Address {

      String city;
      String state;

      Address(String city, String state) {
         this.city = city;
         this.state = state;
      }

      void display() {
         System.out.println("City  : " + city);
         System.out.println("State : " + state);
      }
   }

   public static void main(String[] args) {
```

```
        EmployeeAddressNested emp =
            new EmployeeAddressNested("Sathwika", 1001);

        EmployeeAddressNested.Address address =
            new EmployeeAddressNested.Address("Hyderabad", "Telangana");

        System.out.println("Employee ID   : " + emp.employeeId);
        System.out.println("Employee Name : " + emp.employeeName);

        address.display();
    }
}
```

OUTPUT:

```
C:\Users\Rishitha\Desktop\25EU02070>javac EmployeeAddressNested.java

C:\Users\Rishitha\Desktop\25EU02070>java EmployeeAddressNested
 Employee ID   : 1001
 Employee Name : Sathwika
 City  : Hyderabad
 State : Telangana
```

PROGRAM 14.1 — Student Address

```
class StudentAddressInner {

    int rollNo;
    String name;

    StudentAddressInner(int rollNo, String name) {
        this.rollNo = rollNo;
        this.name = name;
    }

    class Address {

        String city;
        String state;
```

```
    Address(String city, String state) {
        this.city = city;
        this.state = state;
    }

    void display() {
        System.out.println("City  : " + city);
        System.out.println("State : " + state);
    }
  }

  public static void main(String[] args) {

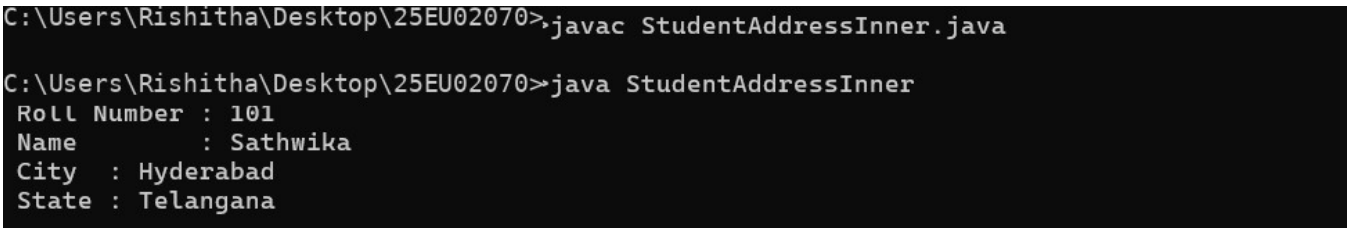
    StudentAddressInner student =
        new StudentAddressInner(101, "Sathwika");

    StudentAddressInner.Address address =
        student.new Address("Hyderabad", "Telangana");

    System.out.println("Roll Number : " + student.rollNo);
    System.out.println("Name        : " + student.name);

    address.display();
  }
}
```

OUTPUT:



PROGRAM 14.2 — Library Management

```
class LibraryBookInner {

    String libraryName = "Central Library";

    class Book {

        int bookId;
        String title;
        String author;

        Book(int bookId, String title, String author) {
            this.bookId = bookId;
            this.title = title;
            this.author = author;
        }

        void display() {

            System.out.println("Library : " + libraryName);
            System.out.println("Book ID : " + bookId);
            System.out.println("Title   : " + title);
            System.out.println("Author  : " + author);
        }
    }

    public static void main(String[] args) {

        LibraryBookInner library =
            new LibraryBookInner();

        LibraryBookInner.Book book =
            library.new Book(
                101,
                "Java Programming",
                "James Gosling"
            );
```

```
        book.display();
    }
}
```

OUTPUT:

